

OUTLIER ANALYSIS

DV2626/DV2599 Machine Learning

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WHAT IS AN OUTLIER?

F. E. Grubbs 1969:

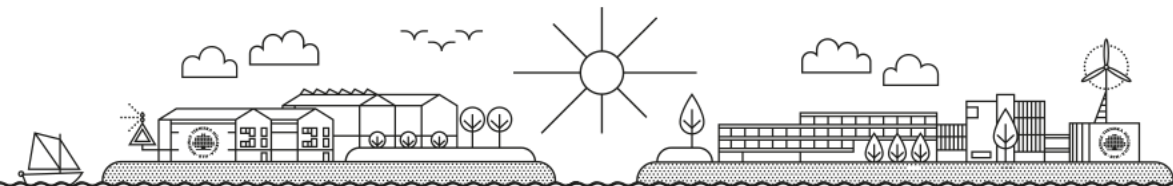
An outlier is an **observation** that **deviates significantly** from other members of the sample in which it occurs.

V. Barnett and T. Lewis 1974:

An **outlier observation** is the one which appears to be **inconsistent** with the remainder of that set of data.

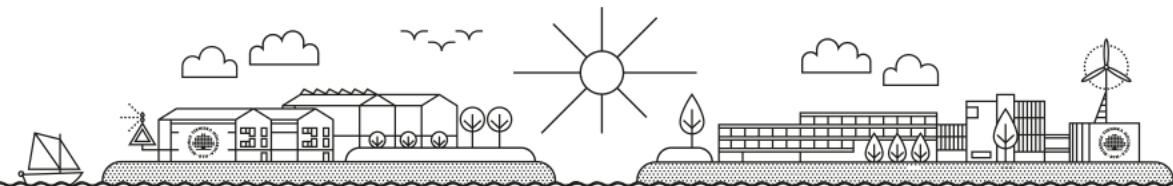
D. M. Hawkins 1980:

An outlier is a **distinct observation** that is seemed to be **generated** by a **different mechanism**.



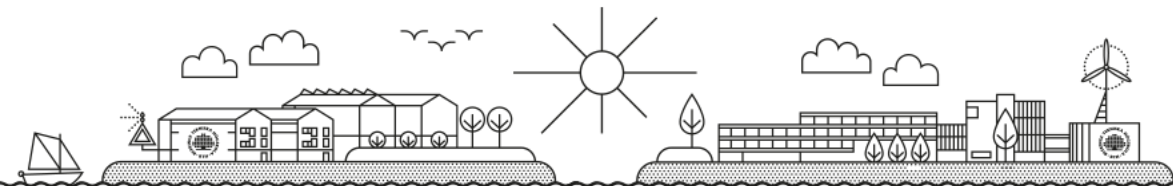
OUTLIER DETECTION

- The problem of finding **patterns** in data that **do not conform** to expected normal behavior
- These distinct patterns are referred to as:
aberrations, anomalies, contaminants, deviations, discordant observations, error, exceptions, noise, novelties, outliers, peculiarities, or surprises
- Example:
 - An anomalous traffic pattern in a computer network -> a hacked computer is sending out sensitive data to an unauthorized destination [Kumar 2005].
 - An anomalous MRI image -> presence of malignant tumors [Spence et al. 2001]
 - Anomalies in credit card transaction data -> credit card or identity theft [Aleskerov et al. 1997],
 - Anomalous readings from a space craft sensor -> a fault in some component of the space craft [Fujimaki et al. 2005].



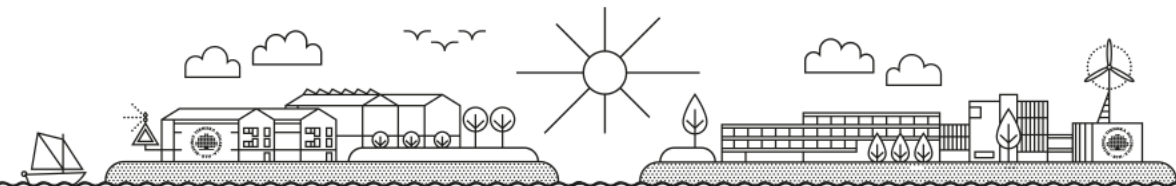
CHALLENGES IN OUTLIER DETECTION PROBLEM

- Defining **normal** and **anomalous** behaviors are difficult and, in many cases, impossible.
- Nature of input data (**data-specific**)
- Availability of the labeled data
- Constraints, and requirements of the application domain problem (**domain-specific**)

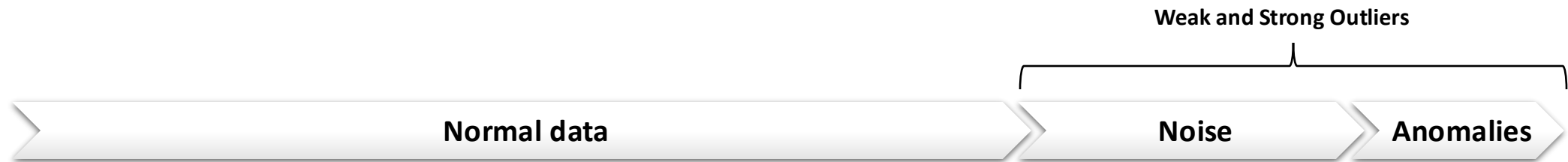


OUTLIER VS. NOISE

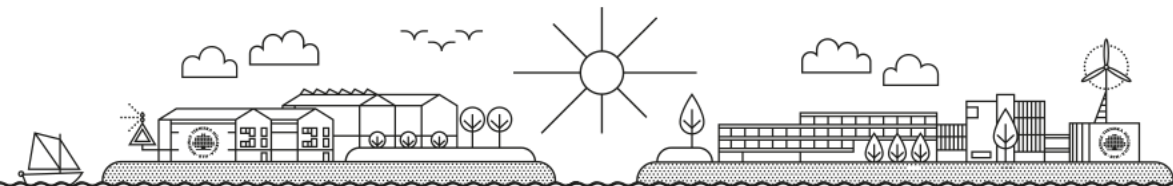
- **Outlier** is related but distinct from *noise*.
- Noise can be defined as a phenomenon in data that is *not of interest* to the analyst but acts as an obstacle to data analysis.
- Type of noise:
 - Attribute noise (e.g., missing values introduced by the measurement tools)
 - Class noise (misabeled instances)
 - A combination of both
- Outliers are considered as *interesting* and/or *unknown patterns* hidden in data; their identification may lead to *new insights, the discovery of system faults, or fraudulent activities*



DATA SPECTRUM FROM NORMAL DATA TO OUTLIERS

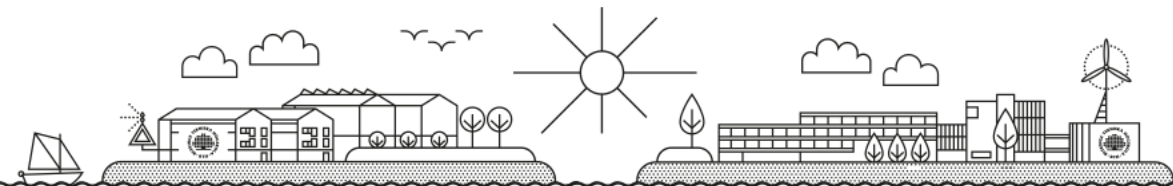


The spectrum of data from normal data to outliers [Aggarwal 2016]



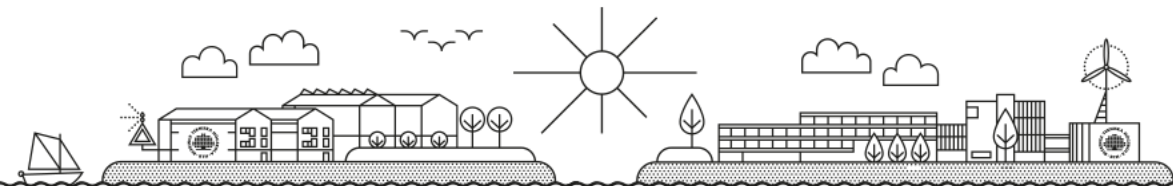
NATURE OF INPUT DATA

- Input data is generally *a collection* of data instances
- Each data instance can be described using a *set of attributes*
- The attributes can be of different types:
 - *binary*
 - *categorical*
 - *continuous*
- The *nature of attributes* determines the applicability of outlier detection techniques.
E.g., Different *statistical* models are required for *continuous* and *categorical* data
E.g., The similarity matrix (pairwise distance between instances) requires techniques that are not dependent on original data instances.



NATURE OF INPUT DATA (CONT.)

- Input data can be categorized based on relationship present among data instances
- Most anomaly detections dealing with point data assume *no relationships* among the data instances.
- Data instances can be related to each other:
 - **Sequence data** -> data instances are *linearly ordered*
E.g., time-series data, genome sequences, and protein sequences
 - **Spatial data** -> each instance is *related* to its neighboring instances
E.g., ecological data, climate data (*spatio-temporal data*)
 - **graph data** -> *vertices*: data instance
edges: connection

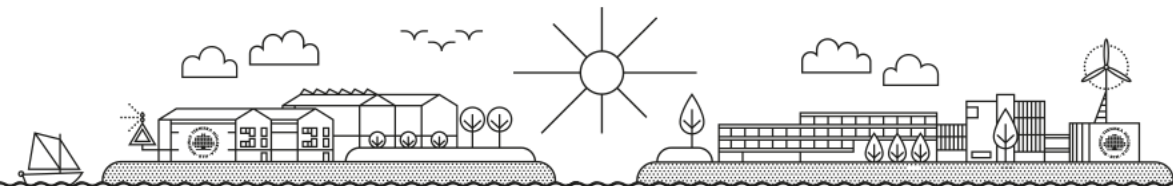


TYPE OF OUTLIERS

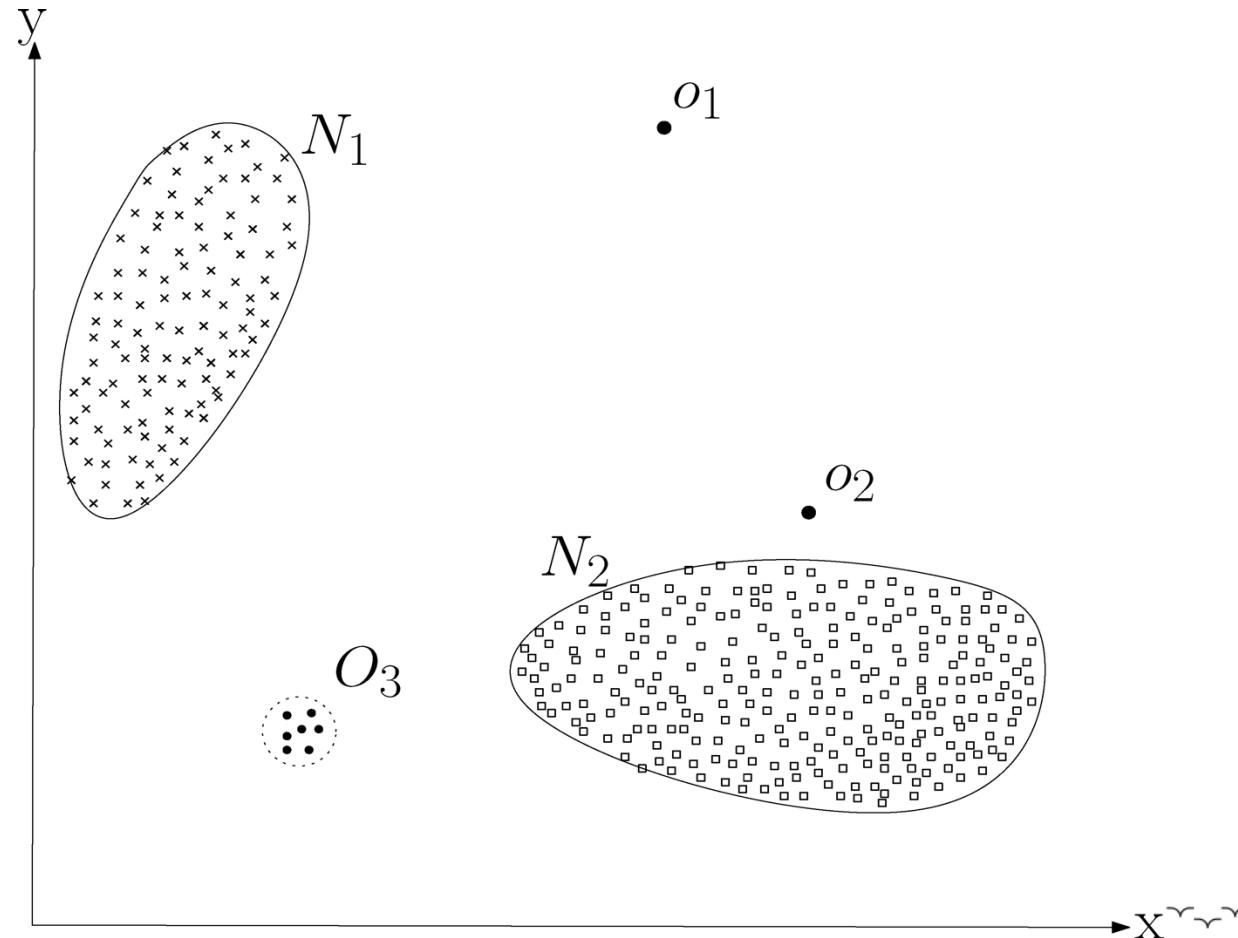
- **Point Outliers:** An individual data point distinct from the entire dataset (global).

E.g., Credit card fraud detection (amount spent)

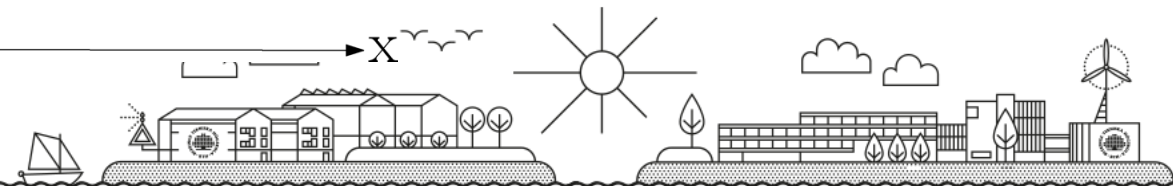
*A transaction with an amount **very higher** than the **normal range** that a person usually spends will be a point anomaly.*



TYPE OF OUTLIERS



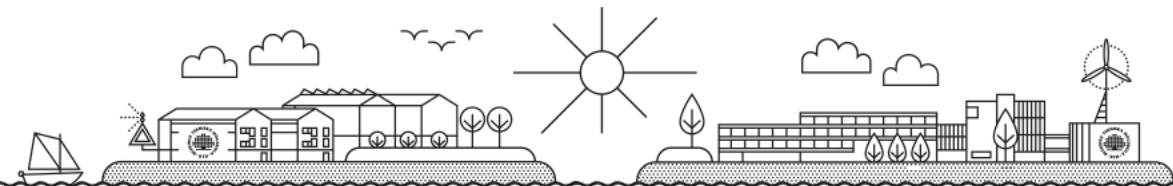
Ref: Chandola et al. 2009



TYPE OF OUTLIERS (CONT.)

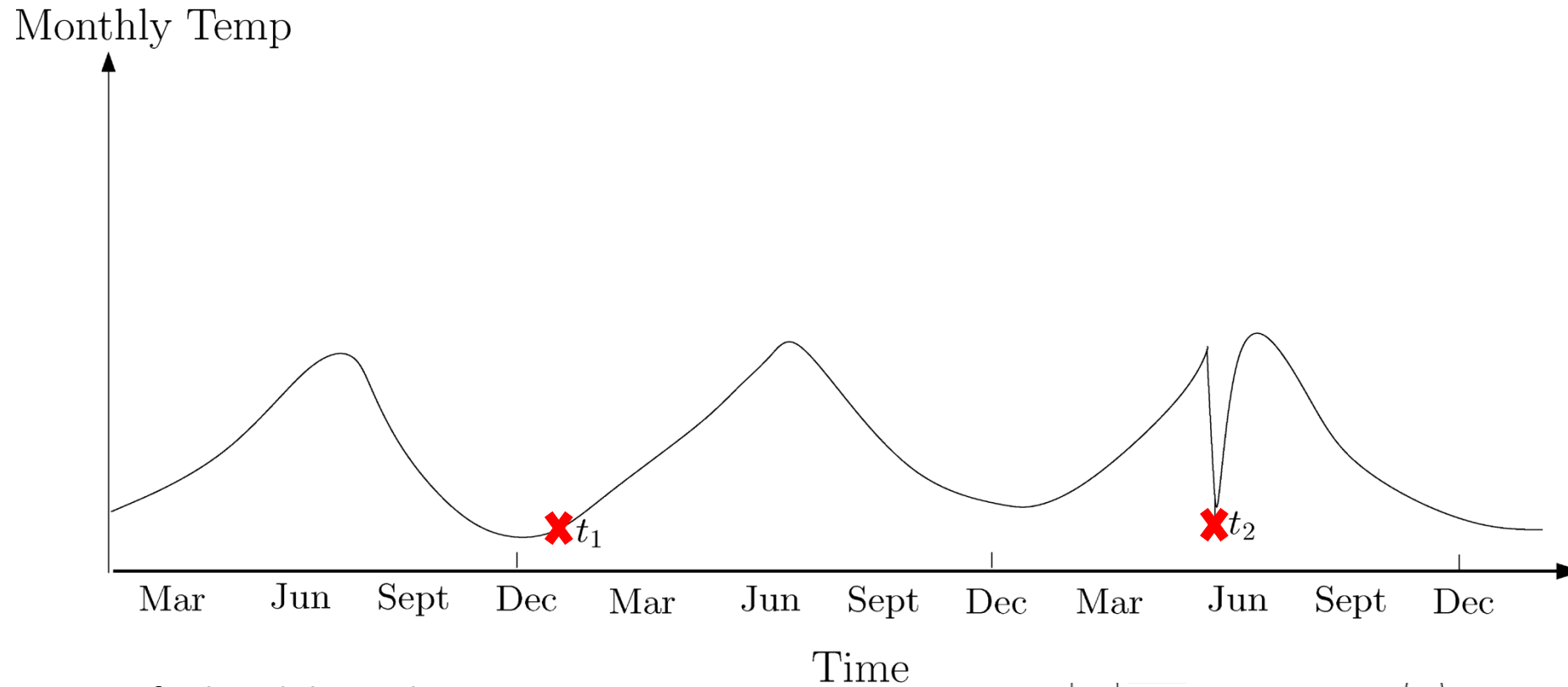
- **Contextual Outliers:** A data point that deviates significantly with respect to a specific context or condition.
- **Context** is related to the structure of the data, and each instance is defined using the following attributes:
 - *Contextual attributes:* determine the context (or neighborhood) for that instance.
E.g., In spatial data sets – (**longitude, latitude**) of a location represents the context
E.g., In time series – **time** is the context
 - Behavioral attributes: define the non-contextual characteristics of an instance.
E.g., Spatial data set describing the average rainfall of the entire world, **the amount of rainfall at any location is a behavioral attribute.**

The anomalous behavior is determined using the values for the behavioral attributes within a specific context.

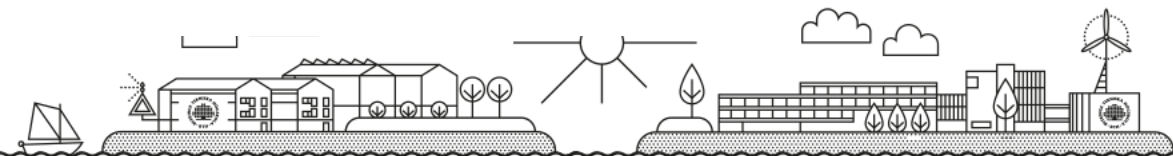


TYPE OF OUTLIERS (CONT.)

Contextual Outliers



Ref: Chandola et al. 2009

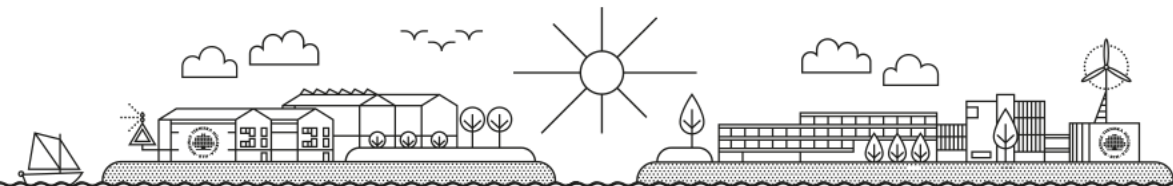


TYPE OF OUTLIERS (CONT.)

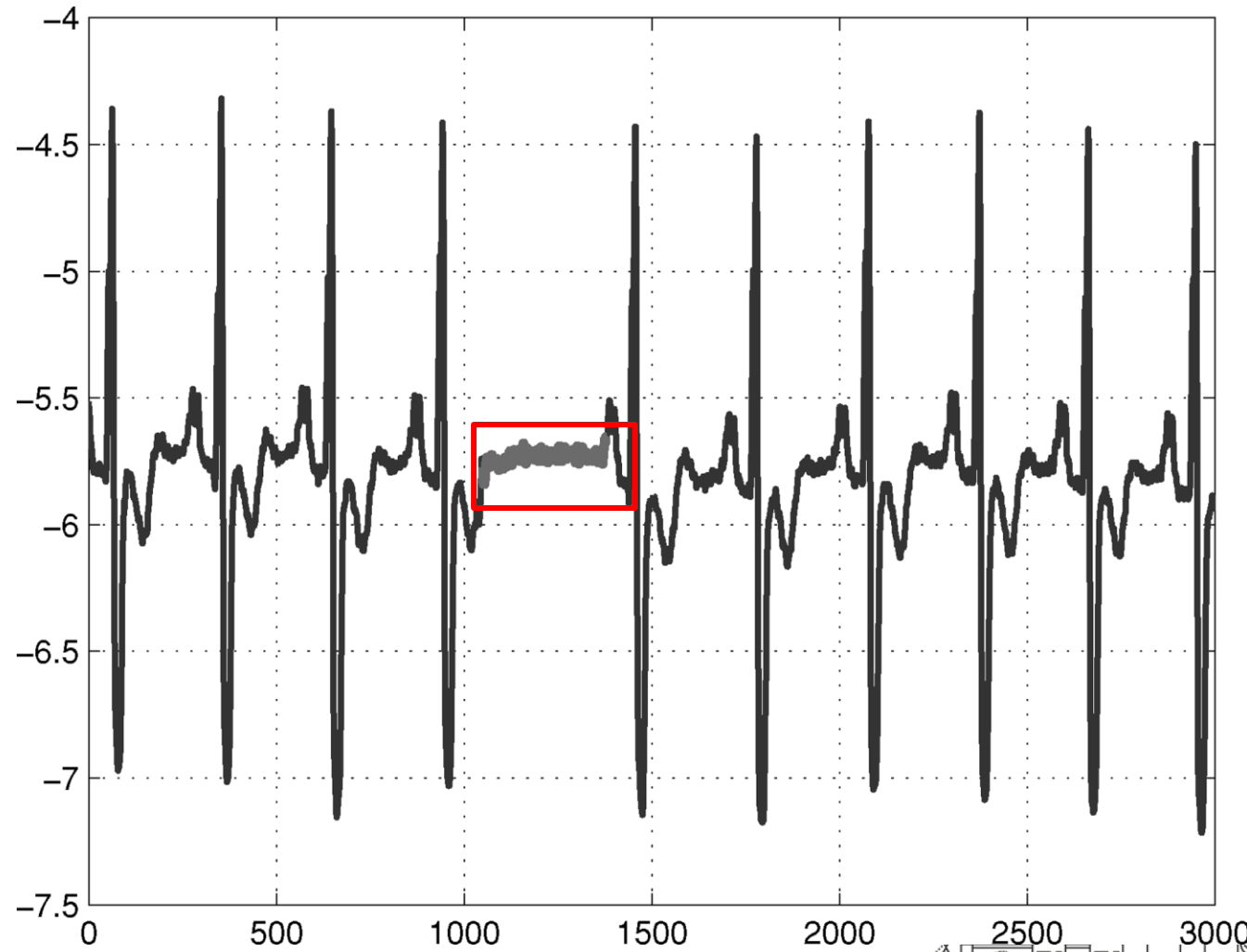
- **Collective Outliers:** A collection (sequence) of related data points that deviates significantly from the entire dataset.

E.g., client roll, created, connectivity changed, bitrate switched, started, program changed, pause, stopped

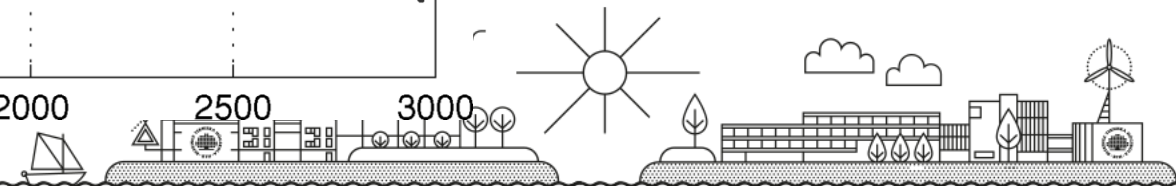
E.g., ..., started, bitrate switched, buffering started, buffering stopped, bitrate switched, buffering started, buffering stopped, bitrate switched, buffering started, buffering stopped, bitrate switched, buffering started, buffering stopped, ...



TYPE OF OUTLIERS (CONT.)

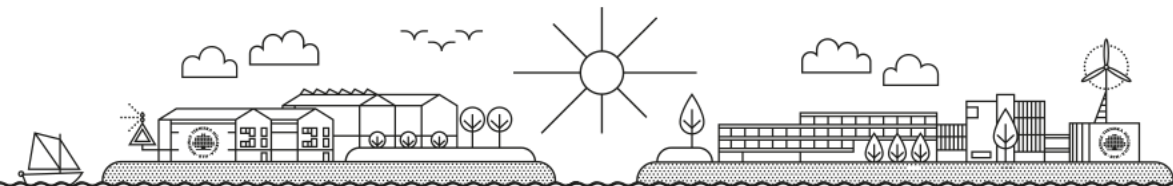


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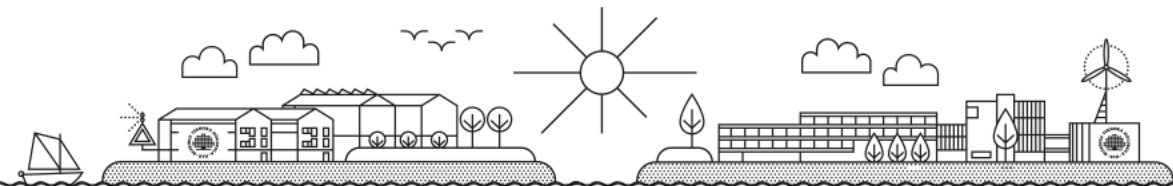
- **Point anomalies** can occur in any data set
- **Collective anomalies** can occur only in sequence data and spatial data.
- **Contextual anomalies** depend on the availability of context attributes in the data, i.e., time series and spatial data.
- A **point anomaly** or a **collective anomaly** can also be a **contextual anomaly** if analyzed with respect to a context.

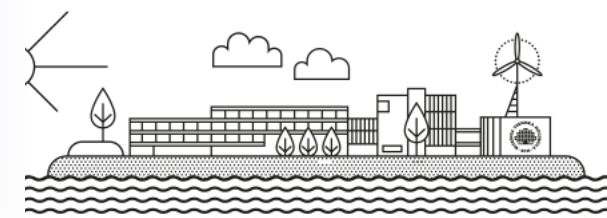


DATA LABELS

○ Presence of labeled data

- Both normality and abnormality are modeled.
- It refers to as *supervised learning*
- *Issue:*
 1. Imbalance data classes





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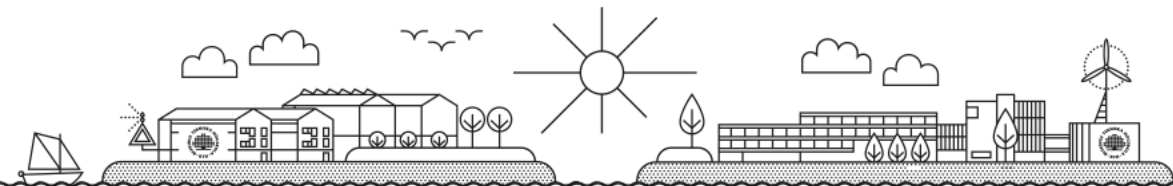
- Both normality and abnormality are modeled.
- It refers to as ***supervised learning***
- *Issue:*

1. Imbalance data classes

Methods:

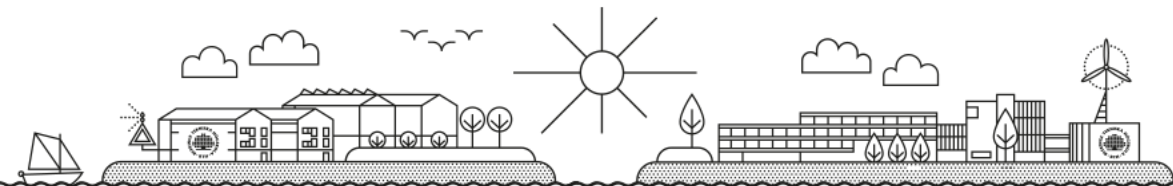
- over-sampling: randomly duplicate examples in the minority class,
- under-sampling: randomly delete examples in the majority class,
- combination of both

2. Obtaining accurate and representative labels, especially for the anomaly class is usually challenging.



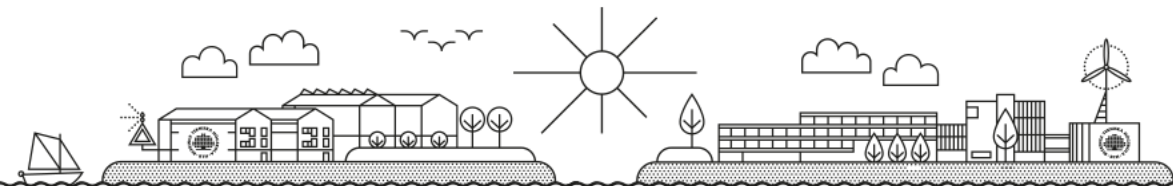
DATA LABELS (CONT.)

- **Define what is normal and only model normality**
 - It is known as **semi-supervised** learning
 - More widely applicable than **supervised** techniques -> there is no need for labels of the **anomaly class**
- **Absence of prior knowledge of the data**
 - It refers to as **unsupervised learning**
 - *Initial assumption*: normal data represents a significant portion of the data and is distinguishable from faults or error
 - *Issue*: If this assumption is not true -> high false alarm rate.



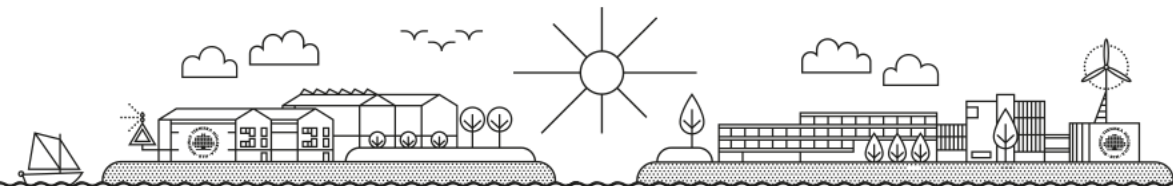
OUTPUT OF OUTLIER DETECTION

- Scores
 - Assign score to each instance
 - Domain expert can use a domain *specific threshold* to select the most relevant anomalies
- Labels (a binary label)



OUTLIER DETECTION APPLICATION

- Fault detection
 - Fault prediction + root cause analysis
- Fraud detection
 - Mobile phone
 - Credit card
 - Insurance claim
 - ...
- Intrusion detection
 - Surveillance
 - Computer networks
- Health anomaly detection
- ...



WHY IS THIS IMPORTANT?

- Outliers can happen because of
 - Mechanical issues,
 - Fraudulent behavior,
 - Human error
- If they are not identified may lead to *uncontrollable* situations



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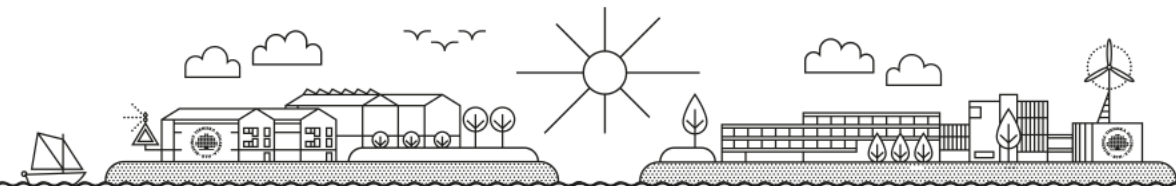
PAPER I



Outlier detection for video session data using sequential pattern mining

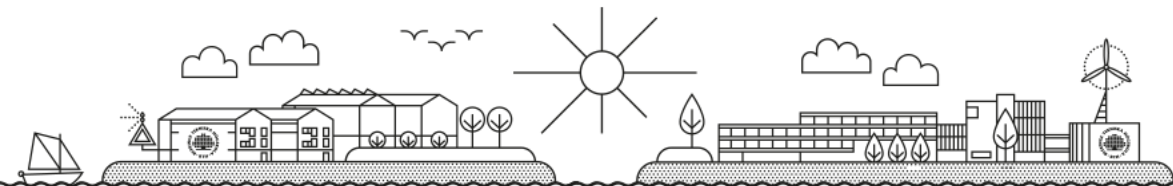
Abghari, S., Boeva, V., Lavesson, N., Grahn, H., Gustafsson, J., &
Shaikh, J.

ACM SIGKDD: Workshop On Outlier Detection De-constructed
2018, London, UK



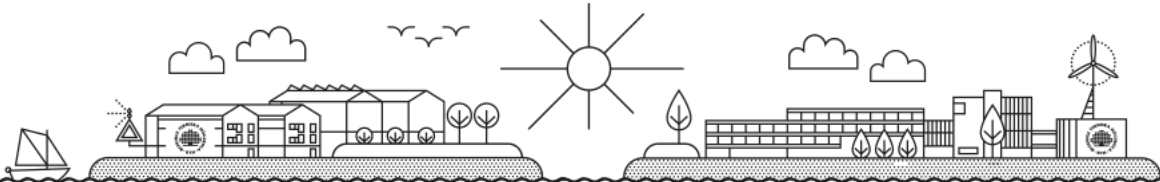
BACKGROUND AND MOTIVATION

- Online video service providers aim:
 - to deliver high quality video contents for viewers.
 - to detect unexpected issues that can highly affect the viewers.
- **Outlier:** finding **unexpected** and **abnormal** patterns in video event sequences.
- **Challenge:** defining a **normal behavior** is difficult without having labeled data.



EXAMPLE OF VIDEO SESSIONS

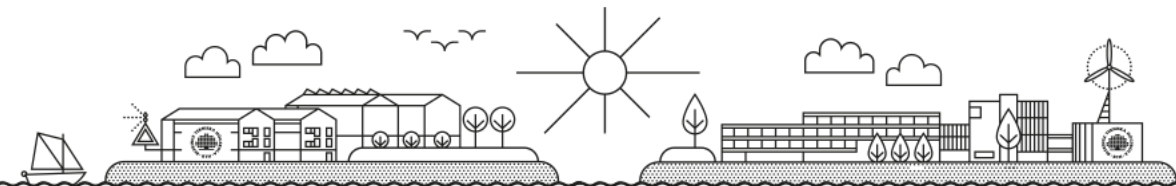
Session ID	Video ID	Date-Time	Event Type	Event ID	Event Type
1	002	Jan-01-17 22:44	client roll	1	bitrate switched
1	002	Jan-01-17 22:45	created	2	buffering started
1	002	Jan-01-17 22:46	connectivity changed	3	buffering stopped
1	002	Jan-01-17 22:47	bitrate switched	4	client roll
1	002	Jan-01-17 22:48	started	5	connectivity changed
1	057	Jan-01-17 22:55	program changed	6	created
1	057	Jan-01-17 23:22	pause	7	error occurred
1	057	Jan-01-17 23:48	stopped	8	paused
				9	play
				10	program changed
				11	scrubbed
				12	started
				13	stopped



EXAMPLE OF VIDEO SESSIONS

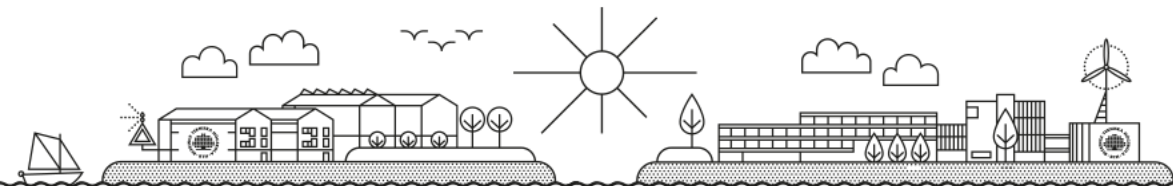
Session ID	Video ID	Date-Time	Event Seq
1	002,057	Jan-01-17 22:44 (64 <i>min</i>)	(4,6,5,1,12,10,8,13)
2	105	Jan-03-17 17:26 (6 <i>min</i>)	(4,6,5,1,1,1,13)

Event ID	Event Type
1	bitrate switched
2	buffering started
3	buffering stopped
4	client roll
5	connectivity changed
6	created
7	error occurred
8	paused
9	play
10	program changed
11	scrubbed
12	started
13	stopped



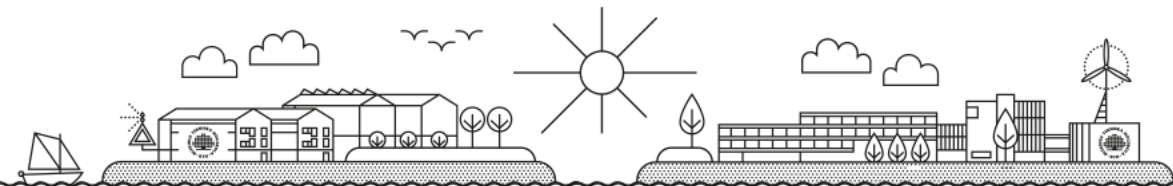
SEQUENTIAL PATTERN MINING AS PART OF THE SOLUTION

- Sequential pattern mining:
 - introduced in 1995 by R. Agrawal
 - finding interesting **frequent ordered** patterns
- Why does it work?
 - Sequences of video events
 - Order of the events is important
 - Any unexpected patterns with high impact on the viewers can be recognized as anomaly.
- E.g., `bitrate switched, buffering started, buffering stopped,`
`bitrate switched, buffering started, buffering stopped`

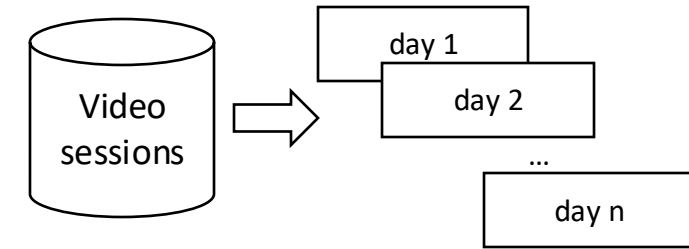


PROPOSED APPROACH

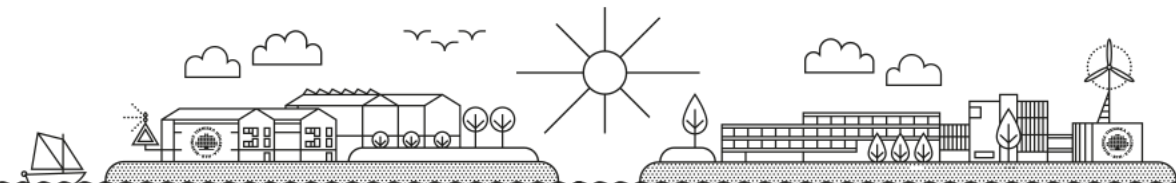
1. Data segmentation
2. Frequent sequential patterns finding
3. Frequent sequential patterns mapping
4. Most frequent & non-most frequent sequential patterns finding
5. *Most frequent sequential patterns* clustering
6. Analysis of *non-most frequent sequential patterns* and outlier detection



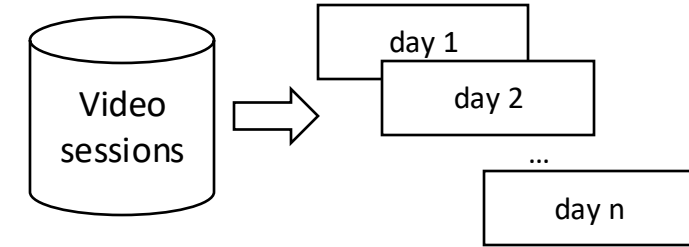
PROPOSED APPROACH (CONT.)



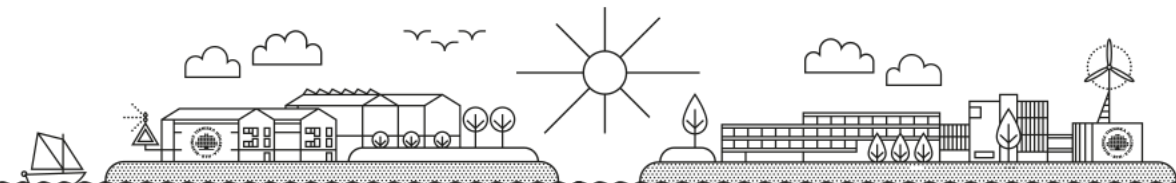
1. Data segmentation
 - Divide video sessions into equal-sized segments
 - Can be performed *hourly, **daily**, weekly, ...*
 - One segment includes all video sessions ***initiated*** at the same time period.
2. Frequent sequential patterns finding
 - Use PrefixSpan algorithm
 - **Input:** (4,6,5,1,12,10,8,13) (4,6,5,8,11,10,8,13)
 - **Output:** (4, 6, 5), (10, 8, 13), (4, 6, 5,10, 8, 13)



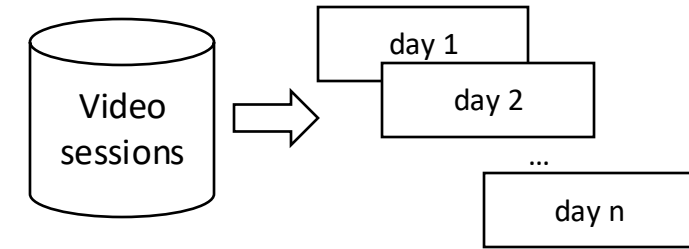
PROPOSED APPROACH (CONT.)



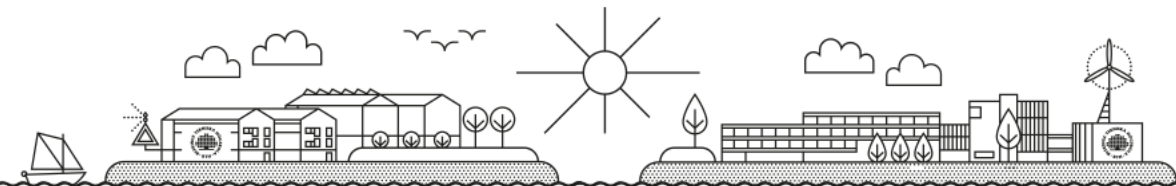
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PROPOSED APPROACH (CONT.)

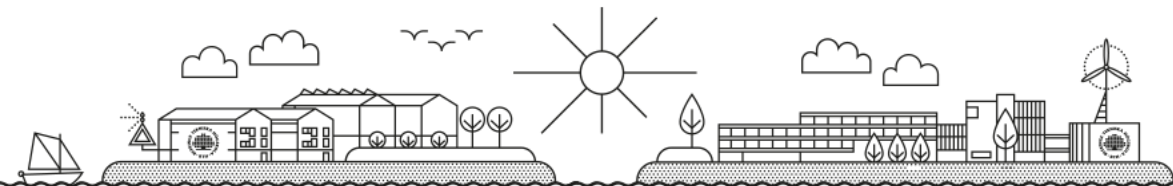


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2. Frequent sequential patterns finding
 - Use PrefixSpan algorithm
 - **Input:** (4,6,5) 1,12,(10,8,13) (4,6,5) 8,11,(10,8,13)
 - **Output:** (4, 6, 5), (10, 8, 13), (4, 6, 5,10, 8, 13)
 - Collective outlier



PROPOSED APPROACH (CONT.)

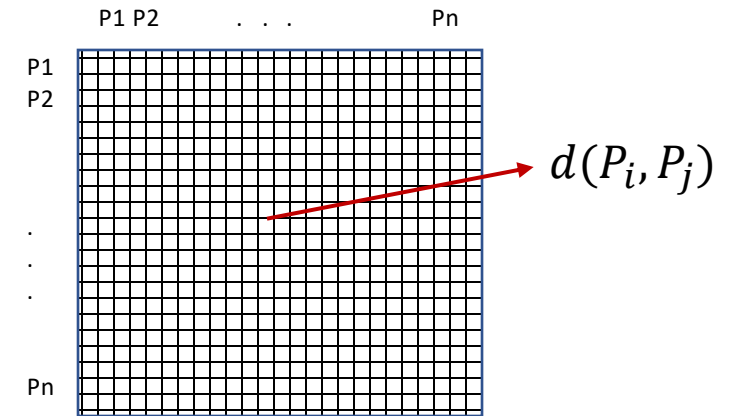
3. Frequent sequential patterns mapping
 - Mark video session that contains a pattern
 - Extract other useful information (e.g., day-of-week, weekend, ...)
 - *Contextual outlier*
4. Most frequent and non-most frequent patterns finding
 - ***Most frequent sequential patterns*** are those that occurred in more than one segment.
 - *Initial assumption*: frequent patterns can be considered as normal.
 - ***Non-most frequent sequential patterns*** can be assumed as potentially unexpected at this stage.



PROPOSED APPROACH (CONT.)

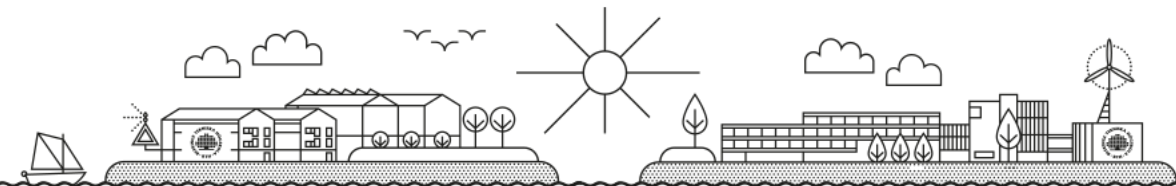
5. *Most frequent sequential patterns* clustering

- *Edit distance* for finding similarity between patterns
 - $d(\text{kitten}, \text{sitting}) = 3$
- k-means
- Affinity Propagation (AP)

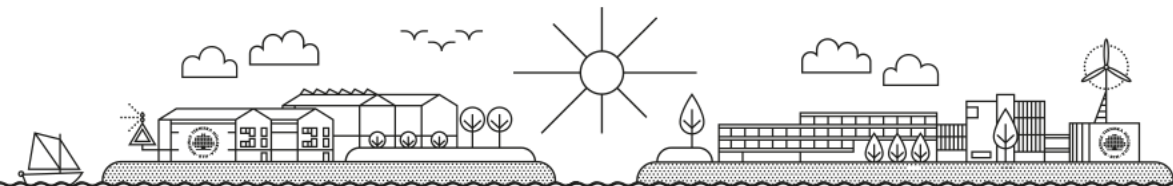
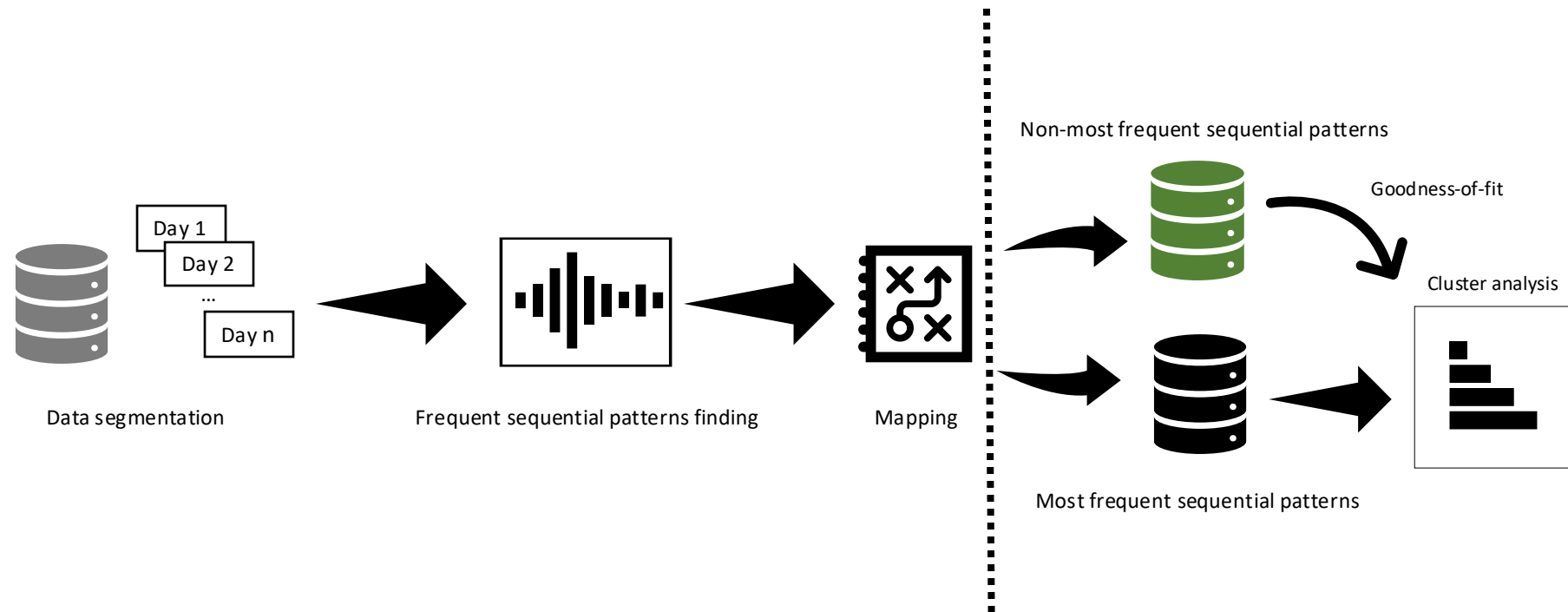


6. Analysis of *non-most frequent sequential patterns* and anomaly detection

- Using the clustering model built in the previous step:
 - Each *non-most frequent sequential pattern* will be assigned to a cluster.
 - The goodness-of-fit of a *non-most frequent sequential pattern* shows whether it is normal or abnormal.
 - The goodness-of-fit can be determined by Silhouette measure $([-1,1])$.

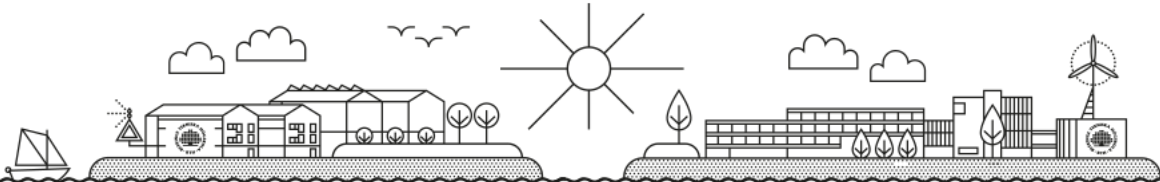


PROPOSED APPROACH



RESULTS: TOP 5 DETECTED *MOST FREQUENT SEQUENTIAL PATTERNS*

Pattern	Oct 2016	Pattern	Nov 2016	Event ID	Event Type
<6, 4, 5, 1, 12, 8, 13>	74,362	<6, 4, 5, 1, 12, 8, 13>	59,340	1	bitrate switched
<6, 4, 5, 1, 12>	12,061	<6, 4, 5, 1, 12>	19,104	2	buffering started
<6, 4, 5, 1, 12, 1>	6,491	<6, 4, 5, 1, 12, 8>	4,449	3	buffering stopped
<6, 4, 5, 1, 12, 8>	5,057	<6, 5, 1, 12, 8, 13>	1,098	4	client roll
<6, 4, 5, 1, 12, 1, 8, 13>	4,025	<6, 5, 1, 12>	503	5	connectivity changed
Total no. of matched Patterns	101,996		84,494	6	created
Total no. of vide sessions	114,407		87,905	7	error occurred
				8	paused
				9	play
				10	program changed
				11	scrubbed
				12	started
				13	stopped

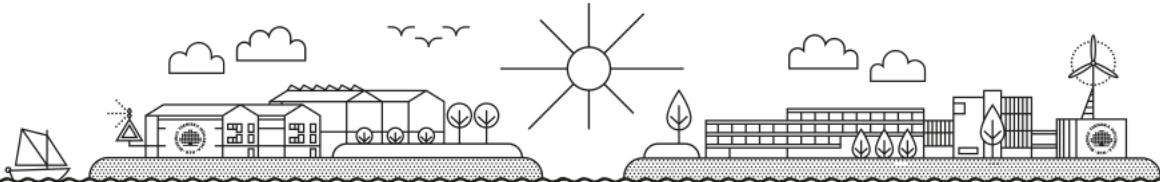


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<i>Pattern</i>	<i>Oct 2016</i>	<i>Pattern</i>	<i>Nov 2016</i>
⟨6, 4, 5, 1, 12, 8, 13⟩	74,362	⟨6, 4, 5, 1, 12, 8, 13⟩	59,340

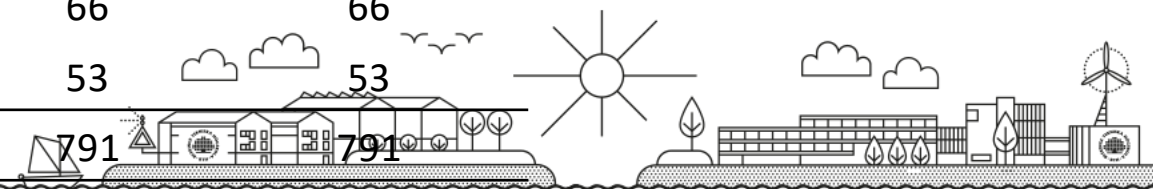
Created, Client roll, Connectivity changed, Bitrate switched, Started, Program changed, Stopped

<i>Total no. of matched Patterns</i>	101,996	84,494
<i>Total no. of vide sessions</i>	114,407	87,905



RESULTS: TOP 5 DETECTED *NON-MOST FREQUENT SEQUENTIAL PATTERNS*

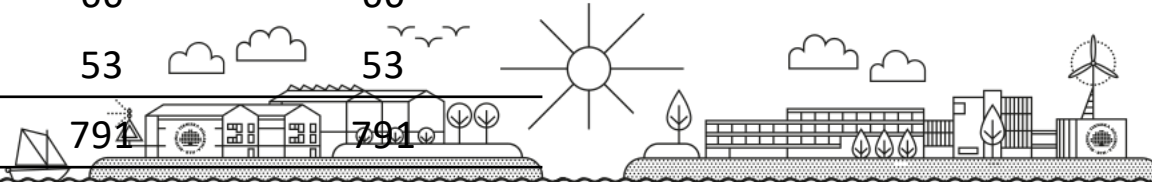
		Affinity Propagation		k-means			
Pattern		Edit Distance	FastDTW	Edit Distance	FastDTW	Event ID	Event Type
Oct 2016	$\langle 1, 2, 1, 1 \rangle$	-	-	-	256	1	bitrate switched
	$\langle 1, 2, 1, 1, 8, 13 \rangle$	101	101	101	101	2	buffering started
	$\langle 1, 12, 10, 8 \rangle$	-	-	148	148	3	buffering stopped
	$\langle 1, 12, 10, 8, 13 \rangle$	136	136	-	-	4	client roll
	$\langle 4, 5, 1, 12, 10 \rangle$	152	152	140	140	5	connectivity changed
Total no. of matched patterns		389	389	389	645	6	created
Nov 2016	$\langle 1, 1, 1, 1, 1 \rangle$	513	513	513	513	7	error occurred
	$\langle 1, 2, 1 \rangle$	92	92	92	92	8	paused
	$\langle 1, 1, 1, 2 \rangle$	67	67	67	67	9	play
	$\langle 3, 8, 13 \rangle$	66	66	66	66	10	program changed
	$\langle 1, 1, 1, 1, 2 \rangle$	53	53	53	53	11	scrubbed
Total no. of matched patterns		791	791	791	791	12	started
						13	stopped



RESULTS: TOP 5 DETECTED *NON-MOST FREQUENT SEQUENTIAL PATTERNS*

		Affinity Propagation		k-means	
Pattern		Edit Distance	FastDTW	Edit Distance	FastDTW
Oct 2016	$\langle 1, 2, 1, 1 \rangle$	-	-	-	256
	<i>Bitrate switched, Rebuffering started, Bitrate switched, Bitrate switched</i>				
	$\langle 1, 3, 1, 12, 10 \rangle$	152	152	140	140
	Total no. of matched patterns	389	389	389	645
Nov 2016	$\langle 1, 1, 1, 1, 1 \rangle$	513	513	513	513
	$\langle 1, 2, 1 \rangle$	92	92	92	92
	$\langle 1, 1, 1, 2 \rangle$	67	67	67	67
	$\langle 3, 8, 13 \rangle$	66	66	66	66
	$\langle 1, 1, 1, 1, 2 \rangle$	53	53	53	53
Total no. of matched patterns		791	791	791	791

Event ID	Event Type
1	bitrate switched
2	buffering started
3	buffering stopped
4	client roll
5	connectivity changed
6	created
7	error occurred
8	paused
9	play
10	program changed
11	scrubbed
12	started
13	stopped



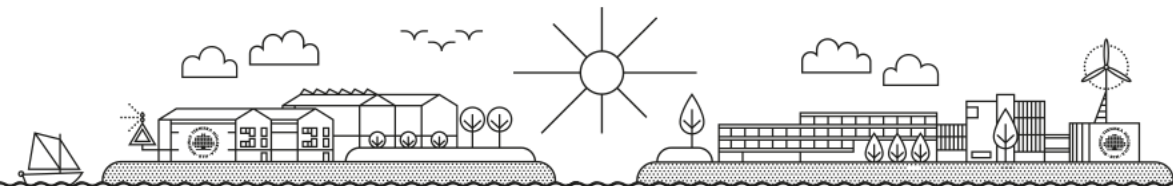
PAPER II



A minimum spanning tree clustering approach for outlier detection in event sequences

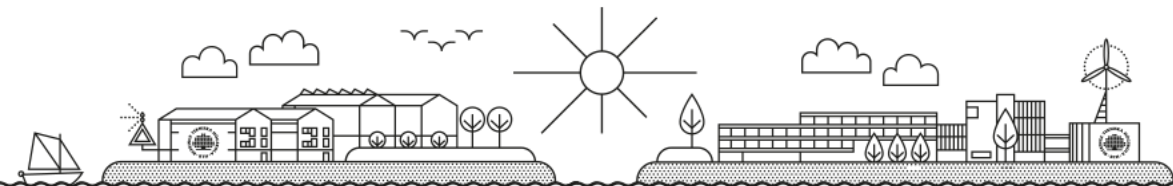
Abghari, S., Boeva, V., Lavesson, N., Grahn, H., Ickin, S., & Gustafsson, J.

In: The 17th IEEE ICMLA: Special Session on Machine Learning
Algorithms, Systems and Applications
2018, Orlando, Florida, USA



BACKGROUND AND MOTIVATION

- Online media streaming and sequence data
- **Individual outlier detection** requires more resources, and it is time consuming
- Instead, **group outlier detection**
 - Similarities
 - Infrequency
 - Distance from majority of data

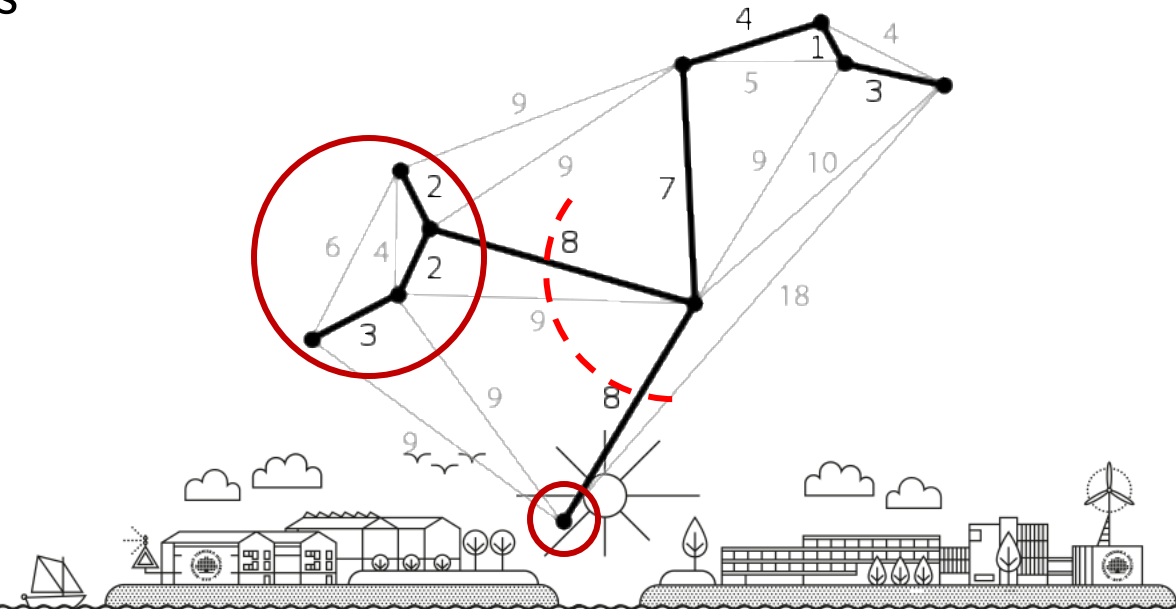


MINIMUM SPANNING TREE AS PART OF THE SOLUTION

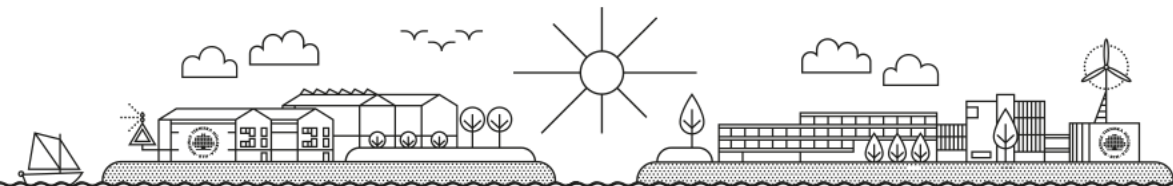
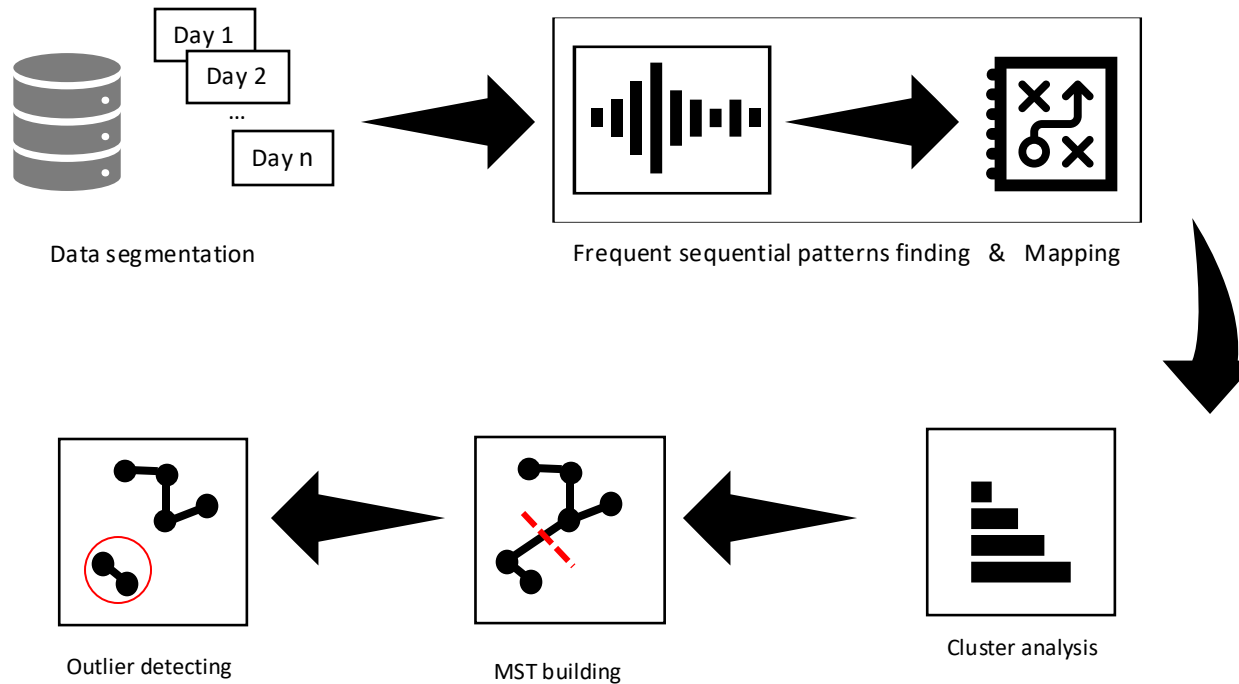
Given an **undirected**, **connected**, and **weighted** graph $G = (V, E)$

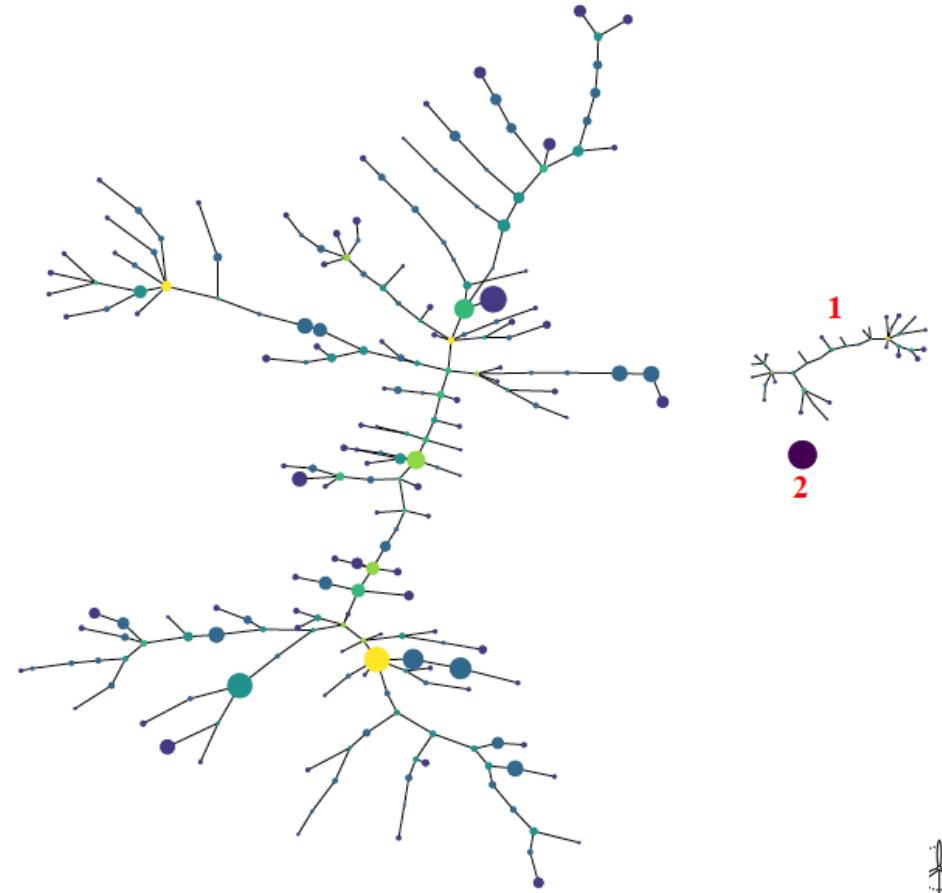
- a spanning tree of the graph G is a tree that covers all the nodes of G
- a tree with the minimum cost of traversing is called the **minimum spanning tree**

- MST is applied on top of the clustering solutions
 - V = exemplars of the clusters
 - E = distances between vertices
- Removing the longest edges
 - Smaller sub-trees are labeled as outliers.



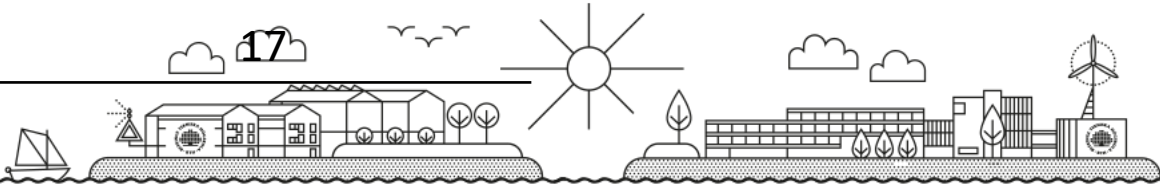
PROPOSED APPROACH





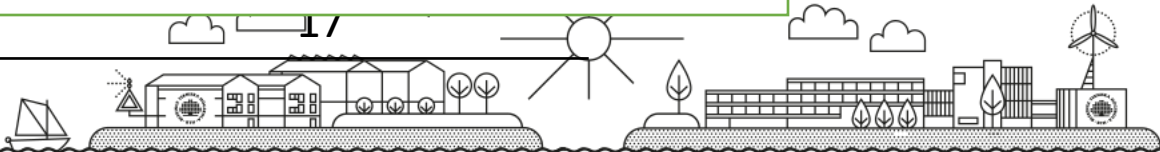
RESULTS: TOP 5 IDENTIFIED SEQUENTIAL PATTERNS SMART METER DATA

	Pattern	Freq. of the pattern
Dataset 1	$\langle 165, 79, 165, 79, 79, 165, 165 \rangle$	27
	$\langle 10, 11, 11, 10, 11, 10, 11 \rangle$	15
	$\langle 20, 22 \rangle$	3
	$\langle 21, 20 \rangle$	3
	$\langle 397, 396, 393, 395, 392, 63, 346 \rangle$	1
Dataset 2	$\langle 20, 20 \rangle$	32
	$\langle 245, 245, 245, 245, 245, 245, 245 \rangle$	31
	$\langle 63, 63, 63, 63, 63, 63, 63 \rangle$	28
	$\langle 21, 21 \rangle$	17
	$\langle 22, 184 \rangle$	17

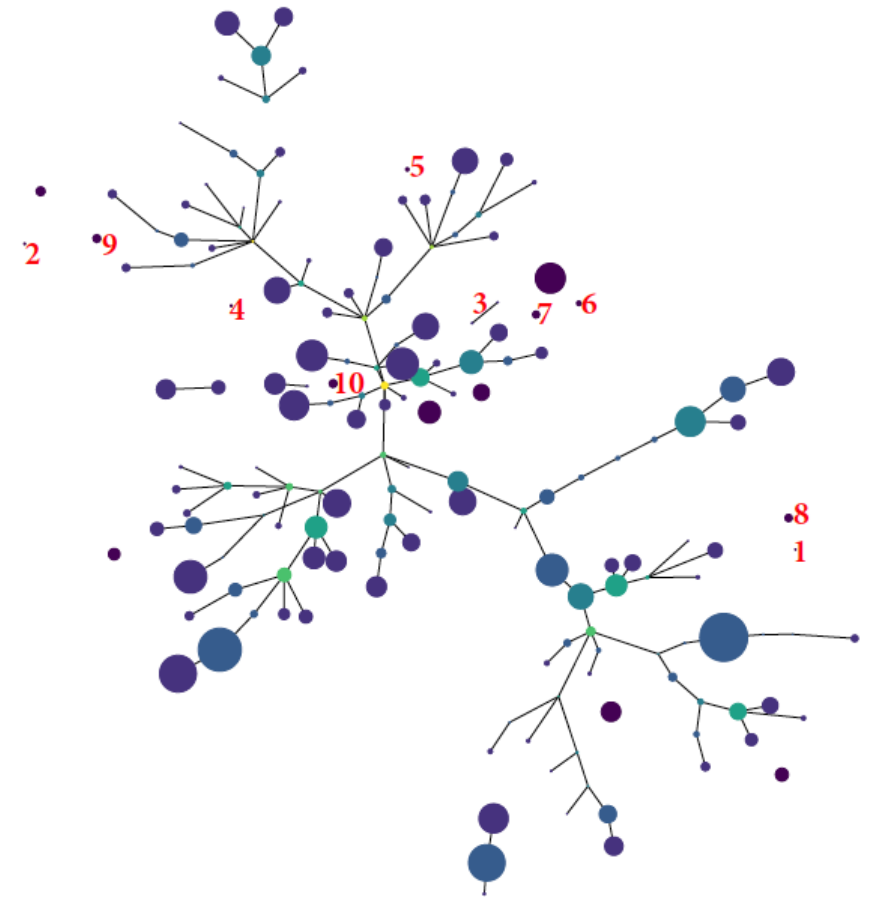
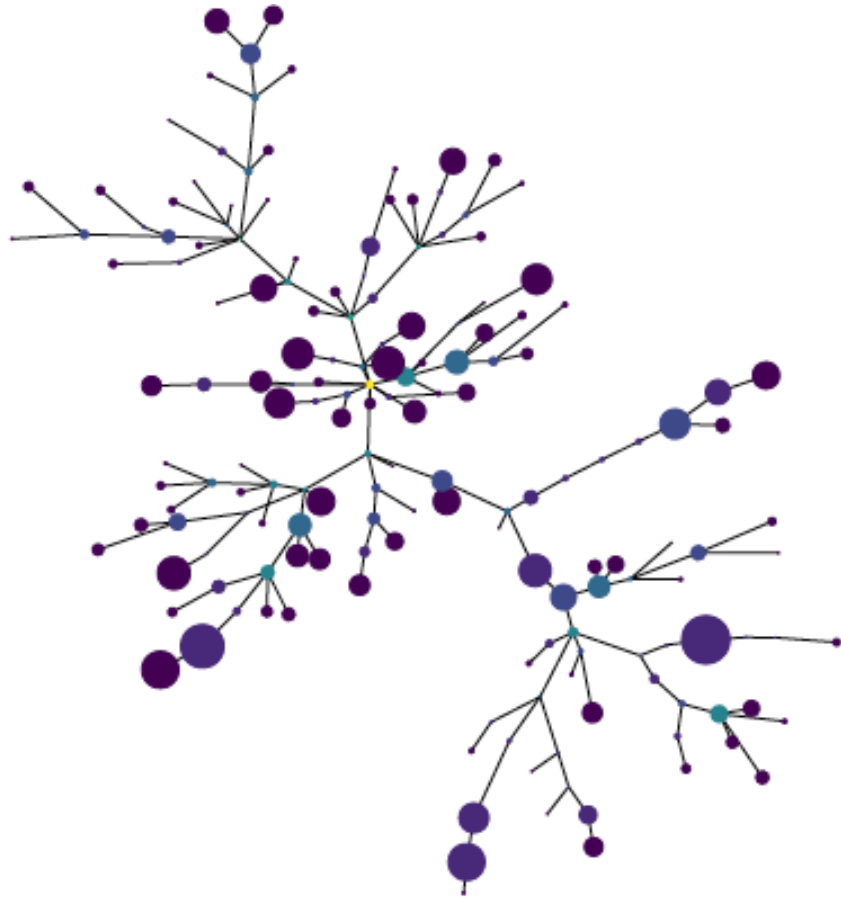


RESULTS: TOP 5 IDENTIFIED SEQUENTIAL PATTERNS SMART METER DATA

	Pattern	Freq. of the pattern
	$\langle 165, 79, 165, 79, 79, 165, 165 \rangle$	27
Remote communication module OK, Communication board access error, PLC or GSM/GPRS Module		
D	$\langle 21,20 \rangle$	3
	$\langle 397, 396, 393, 395, 392, 63, 346 \rangle$	1
	$\langle 20,20 \rangle$	32
2	$\langle 245, 245, 245, 245, 245, 245, 245 \rangle$	31
E Meter command error, E Meter command error		
	$\langle 22,184 \rangle$	17



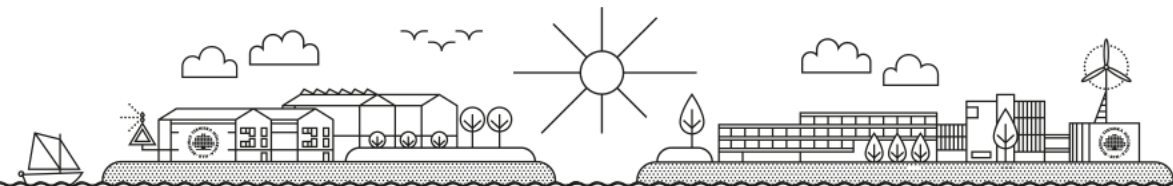
RESULTS: MEDIA DATA



RESULTS: VIDEO SESSION DATA

- Validation of the results
 - Video session data from February 1, 2018
 - Randomly selected 9 normal + 9 outlier video sessions
 - The labels of the data were unknown to the expert
 - 12 out of 18 were detected correctly

- Experts' comments concerning some of the sessions
 - V1. "Short session (15,5 sec), no buffering, good bitrate, hard to tell, I tend to OK."
 - V2. "No buffering, rather short, 10 sec, bitrate rather good. OK."



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